

Supplementary Material for “Auditing Smart-Home Agents: Benchmark Repetition and Grounded Visual Control”

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1 SCOPE AND ARTIFACT POLICY

This supplement specifies public-data construction, model interfaces, integrity checks, and human validation. It contains no estimated model results. The accompanying anonymous artifact contains the public generation code, frozen manifests, raw outputs, evaluators, attestations, and aggregate reports needed to verify the reported results; source images remain under upstream terms.

SHA-256 binds each public run to its manifest, predictions, evaluator, and configuration. Reporting requires exactly one error-free prediction per task; manuscript export additionally requires an explicit registry match.

2 FROZEN PUBLIC PROTOCOLS

2.1 VisualHomeBench-Public-v2

The source is the HomeObjects-3K v1.0.0 held-out validation partition. We retain the original image pixels and YOLO-format boxes. Five source classes are mapped to a zero-argument smart-home API:

Table 1: Frozen public class–operation mapping.

Source class	Allowed operation
lamp	turn_on
television (tv)	turn_on
laptop	wake
window	open
door	open

For each source image, deterministic generation constructs: (i) an explicit request naming a visible class and operation; (ii) a paired implicit request expressing the desired state; (iii) a relational request when a uniquely named anchor and at least two target instances permit deterministic nearest-center selection; and (iv) an infeasible request for an actionable class absent from the image. Repeated positive instances use the largest box unless the request is relational. Negative requests are split equally between explicit and implicit wording. The full manifest contains 1,521 tasks over all 404 validation images; its digest is `ccfa52bfd9499c8df11bba7ff96d5e36a5c8bcb24f8ef16baf2434d8a25ff30e`.

Because the 170-task core informed development, a prospectively frozen view excludes all tasks from its 110 images: 992 tasks

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over 294 images, digest `393d248f101234b2e249b70e78873f6c8cfc54324b098494ae56e66542bd7c9a`. It is a robustness analysis; the complete manifest measures benchmark coverage.

An independent audit reopens every image and label file and verifies dimensions, source-box membership, largest and nearest-anchor selection, paired scene identity, negative-target absence, operation legality, sequential identifiers, and unique negative images. The audit passes with zero violations.

2.2 ActionContrast-v1

The main visual suite exposes one operation per class. ActionContrast-v1 therefore fixes the image, source box, and class while contrasting opposite operations: open/close for doors and windows, and turn_on/turn_off for lamps and televisions. Each of 40 target-image groups yields two operations in explicit and implicit forms, giving 160 tasks and 80 paired comparisons. The manifest digest is `22f868a3eada7085a5e4a8a66a51efce07a1dab6b47b4eba8133c0f4a8993989`. Laptop is excluded because only three held-out images contain one.

A prediction-blind visual audit flagged two apparent source-label mismatches. Published labels remain the primary analysis. A separately named 152-task drop-only sensitivity set is preliminary until independent adjudication and cannot replace the primary result.

2.3 Multi-v1

Multi-v1 contains 48 explicit/implicit pairs over 48 non-reused images. Six two-class combinations among door, lamp, television, and window contribute eight scenes each. Success requires exactly two schema-valid calls, both operations and both boxes correct at $\text{IoU} \geq 0.5$, and no extra call. Call order is ignored; mean subgoal success is descriptive and cannot convert partial completion into exact-set success. The manifest digest is `3a1990562a3cfc7d585162d203a27b47e7d71f32a82bebe1306fad35e88a52d4`.

3 MODEL INTERFACES

3.1 Direct visual model

The task record supplies the same schema instruction to every direct model. The user message concatenates this instruction, original image dimensions, and the request, followed by the unmodified image:

```
Return JSON only with keys valid, target_class, bbox, and operation.
bbox must be [x1,y1,x2,y2] integer pixel coordinates in the original
image. The available device API is: lamp:turn_on, tv:turn_on,
laptop:wake, window:open, door:open. Use only these exact class and
operation strings. If no device satisfying the request is visible,
set valid=false and set target_class, bbox, and operation to null.
Image dimensions are <width>x<height>.
Request: <request>
```

The reproducible local baseline is Qwen2.5-VL-3B-Instruct Q4_K_M through an OpenAI-compatible llama.cpp server. Decoding uses temperature 0, seed 42, one parallel request, a 6,144-token context, and at most 128 completion tokens (256 for the two-call Multi-v1 output). The model and projection digests are, respectively, d02fe9b69ad8cadbbd228e387667af66612c44bed29ffc8eb1e7caf9ac486c12 and 980c9b2f78c04e6cff93d277ada09e768394f112d75db3b4e9dea8a69f9fb904.

llama.cpp intermittently returned projector bad-allocation errors. Failed attempts are preserved and excluded. After 686 clean rows, an audited controller restarts the identically configured server every four requests, aborts after three failures per task, and records hashes. Latency is not compared.

3.2 Controlled 7B/8B sweep

The cross-model diagnostic evaluates Qwen2.5-VL-7B-Instruct Q4_K_M and InternVL3-8B-Instruct Q4_K_M on all 1,521 tasks. Both use the same llama.cpp bundle, temperature 0, seed 42, one request at a time, a 6,144-token context, 128 completion tokens, and 15 GPU layers. Source images are not modified on disk. For resource-normalized requests, each image is deterministically re-encoded in memory as JPEG quality 95 after Lanczos resizing to a maximum side of 768 pixels; every prediction records both source and encoded-image SHA-256 digests. This policy differs from the original full-resolution 3B run, so the sweep is a controlled failure diagnostic rather than a size ranking.

The Qwen run contains 1,521 unique, error-free outputs and has prediction digest 344a04909ecadb19226c6bb070c5c16dc36e92a604475c124aa0e5e6e7ed8484. The InternVL run likewise contains 1,521 unique, error-free outputs and has prediction digest 71356ae61b1a7eb25e716b240e49edc4bbff7bffb5ec18a9f5b35724ba289545. Attestations additionally bind model, projection, configuration, server, runner, and event-log hashes. InternVL resumed once after an outer shell lease expired at 136 clean rows; the existing unique rows were hash-checked and no transport recovery occurred.

The shared-encoding prompt asks for source-image pixel coordinates even though the model observes a deterministically resized image; the Qwen2.5-VL model documentation specifies actual-image-size coordinates rather than a normalized grid. In a post-hoc frame audit, 959 of Qwen-7B’s 1,075 valid boxes (89.2%) fall wholly inside the encoded dimensions. A deterministic sensitivity multiplies each Qwen box by the recorded source-to-encoded width and height ratios while preserving its validity, class, and operation. Full success changes from 19.7% strict raw to 31.1% [28.4, 33.9], and G@.5 from 0.5% to 18.4%. The identical transform reduces full success for InternVL3 (17.4% to 16.6%) and for the alias-repaired GPT control (41.5% to 24.8%); their outputs are therefore not changed. This is a model-specific post-hoc sensitivity rather than per-example frame selection.

3.3 Proprietary frontier API control

GPT-5.6 Sol is evaluated through the Responses API on the same 1,521 tasks and the same deterministic 768-pixel JPEG-95 request encodings used by the 7B/8B sweep. The request retains the benchmark prompt verbatim and sets medium reasoning, low response

verbosity, a 2,048-token output ceiling, explicit high image detail, four bounded concurrent requests, no tools, no structured-output enforcement, and store=false. Temperature and seed are not exposed for this configuration. A prediction-blind pilot covers one task from each of 16 task-type–target strata and checks only API access, parsing, token sufficiency, latency, and projected cost. A 512-token preflight that exhausted its budget on reasoning is preserved; the prompt was not changed. The strict scorer preserves schema violations such as class-prefixed operation strings. GPT-5.6 Sol produced this exact alias on 1,175 of 1,521 tasks after the prompt displayed the API as, e.g., door : open, while separately asking for operation: "open". We therefore report both the strict raw score and a minimal alias-only repair that maps class:operation to the disclosed operation token. It changes no validity decision, class, or geometry; its task-level output and implementation are retained separately from the more general validator.

The GPT-5.6 Sol API run contains 1,521 unique, error-free outputs and 1,521 unique provider response identifiers. Its prediction digest is d04714a06b10a69433ed7557f61f86d60f5cbf63cb75508da82b08f599bc1a96. The frozen configuration digest is f393dc8756d7e31e01711029369db14d24bffb5c550e1c1911686455948ec260. The attestation records 903,154 input, 279,211 output, and 231,698 reasoning tokens; the standard-rate upper-bound estimate is \$12.89. No credential or secret material is retained.

3.4 Grounded planners and validator

Oracle and detector-inventory planners receive candidate triples (candidate_id, class, bbox) from source labels or YOLO11n. The model selects identifiers and operations but cannot rewrite grounding-owned class or geometry. Oracle inventory is an annotation diagnostic, not a deployable system.

The deterministic validator canonicalizes disclosed aliases, clips finite coordinates, derives the registered operation, and rejects illegal classes or degenerate boxes without inspecting ground truth. Passed, repaired, rejected, and unparsed counts are retained.

3.5 Targeted anomaly audit

We audited the door and relational slices after observing near-zero scores for the 3B direct and oracle-inventory planners. A perfect-prediction loopback and a source-label inventory with the corrected frozen rules each score 100% overall, on the 133 positive door tasks, and on all 171 relational tasks. This shows that the manifest and evaluator admit the intended answers. The audit did find that the original deterministic rule parser could choose an actionable anchor (e.g., window) rather than the requested noun (e.g., door) in “door nearest to window.” A target-before-relation parser and regression test correct this defect; all reported rule results use the frozen corrected file.

To distinguish prompt, runtime, and scoring failures, we then froze an outcome-blind 80-task sensitivity sample: the first 20 explicit-door and first 20 implicit-door tasks by identifier, plus 40 relational tasks selected by a deterministic class round-robin. Table 3 reports only complete runs on this same task set. The clarified contract states that an operation must be the suffix alone and that in “X nearest to Y,” X is the target and Y the anchor.

Table 2: Post-audit checks on anomalous slices. FR denotes false rejection.

System/check	Door	Relational
Source labels + corrected rules, success	100.0%	100.0%
Qwen-3B direct, FR count	126/133	168/171
Qwen-3B oracle inventory, FR count	131/133	169/171
GPT-5.6 Sol + alias repair, success	51.9%	24.6%
Detector + corrected rules, success	–	47.4%

The oracle arm changes only this wording. The 7B runtime arm retains the frozen prompt and changes only llama.cpp’s minimum visual-token allocation from the earlier run’s observed 640-token ceiling to 1,024 image tokens.

Table 3: Post-hoc sensitivity on the frozen 80-task anomaly sample (percent). All tasks are positive; these rows do not replace primary results.

Setting	Full	G@.5	Door	Relational
Qwen-3B direct, frozen prompt	0.0	0.0	0.0	0.0
Qwen-3B direct, clarified contract	2.5	2.5	0.0	5.0
Qwen-3B oracle inventory, frozen prompt	0.0	0.0	0.0	0.0
Qwen-3B oracle inventory, clarified contract	0.0	0.0	0.0	0.0
Qwen-7B, frozen runtime, strict coordinates	0.0	0.0	0.0	0.0
Qwen-7B, frozen runtime, encoded-frame rescale	11.2	12.5	13.5	7.5
Qwen-7B, 1,024 image-token minimum, strict coordinates	3.8	3.8	5.8	2.5

The sensitivity rejects a single malformed-prompt explanation. Clarifying the contract recovers only 2/80 direct successes and no oracle-inventory success; the latter still rejects 78/80 positive requests. Raising Qwen-7B’s visual budget recovers 3/80 strict successes, whereas remapping the earlier run’s encoded-frame boxes recovers 9/80. The higher-budget run places only 15/59 valid boxes wholly inside the encoded dimensions and is therefore not given that rescale. Thus runtime and coordinate conventions explain part, but not most, of the anomaly. These post-hoc results narrow the claim to the tested Qwen configurations; they do not support a claim that all VLMs fail these slices. The frontier result also separates semantic performance from serialization: strict raw full success is 18.6%, while changing only the class-prefixed operation alias raises it to 41.5% [38.6, 44.4]. The artifact file `anomalous_slice_audit.json` contains input hashes and the first ten positive door and first ten relational cases by frozen task identifier, including instructions, truths, predictions, and raw responses.

4 SCORING AND STATISTICAL PROTOCOL

Positive full success requires a feasible prediction, correct class and operation, and IoU at least 0.5. Negative success requires the canonical strict rejection with all call fields null. Balanced success is the unweighted mean of positive full success and infeasible rejection. Micro results are accompanied by an unweighted macro average over positive source classes because class frequencies are unequal.

Input validation rejects missing, duplicate, or unexpected task identifiers and any recorded transport error. Confidence intervals use 5,000 nonparametric cluster-bootstrap draws over source images, preserving dependence among queries derived from the same image. The explicit–implicit contrast pairs each target’s requests and jointly flips all differences from one image in a

two-sided 20,000-draw Monte Carlo sign-flip test (seed 20260721); its interval likewise resamples images. Pair-level McNemar counts are descriptive. This wording contrast is the pre-specified inferential comparison; class, box-size, object-density, multiplicity, relational, and failure-taxonomy slices are descriptive.

Tiny-class descriptive result. Laptop is omitted from the main slice table because it has only seven positive examples. Descriptively, direct G@.5, direct full success, validated full success, and oracle-inventory full success are 42.9%, 14.3%, 42.9%, and 14.3%, respectively; these unstable point estimates do not support class-level conclusions.

Failures are assigned once in a fixed priority order: unparsed output, schema violation, unsafe false acceptance, false rejection, wrong device, wrong operation, grounding failure, and success. Qualitative examples use the first task identifier available in a fixed taxonomy order, preventing score-based manual selection.

5 HUMAN VALIDATION OF IMPLICIT REQUESTS

A deterministic, prediction-blind packet contains 200 of 675 eligible implicit requests, stratified across validity, requested class, and available target-size bins. Each of three fluent English speakers sees only a re-encoded public image and request. They rate naturalness (1–5), whether a state change is implied, the intended target and operation, and whether the scene uniquely supports the target. A worst-case finite-population calculation over the frozen strata gives a 7.457-point half-width at 95%; this is a design bound, not an estimated support rate.

An item is independently supported when at least two raters assign naturalness at least 3 and agree with the intended target, operation, and scene support. For a positive request this requires the labeled class/operation and scene support; for an infeasible request it requires the requested class/operation and a judgment that the scene does not support it. The frozen questionnaire asked for linguistic intent separately from scene support, and all three raters used this interpretation. We documented this post-collection coding clarification before manuscript export and did not edit any judgment. Reporting includes supported count, a finite-population stratified 95% interval, median and interquartile naturalness, majority availability, and nominal Krippendorff agreement for target, operation, and scene support. The analyzer requires three distinct rater codes, exact 200-item coverage in each file, schema-valid responses, and current input hashes. Because the round-robin sample deliberately overrepresents rare strata, the population support estimate post-stratifies validity/class/size-specific rates to all 675 eligible tasks; the unweighted packet rate is retained separately. The frozen primary manifest is never edited from these judgments; any filtered sensitivity result is separately labeled.

6 MINIMAL REPRODUCTION SEQUENCE

From the artifact root, the public audits and tests are invoked as follows:

```
.venv/Scripts/python experiments/audit_homebench.py
.venv/Scripts/python experiments/audit_homebench_template_overlap.py
.venv/Scripts/python experiments/audit_visualhomebench.py
.venv/Scripts/python experiments/audit_action_contrast.py
.venv/Scripts/python experiments/audit_multi_device.py
.venv/Scripts/python -m unittest discover -s experiments -p "test_*.py" -v
```

Export requires matching manifest, prediction, evaluator, and registry hashes. Attestation also binds the model, projector, server, configuration, controller, and recovery log.